USN											

24CS320

JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Computer Science and Engineering

III Semester BE: **Test-1**

Digital System Design

Duration: 1 Hr. 15 Min. **Date:**11.11.2025 **Max. Marks:** 30

NOTE: Question No. 1 is compulsory, answer either Question No. 2 or 3 completely.

Q.NO	CO	CD	PI	QUESTION	MARKS
1(a).	1	L3	2.2.3	Using the Quine Mc Clusky method. Find the Prime Implicants and Essential Prime Implicants for the given Boolean function. F (A, B, C, D) = Σm(0, 1, 2, 8,10,11,14,15)	10
1(b).	1	L3	2.2.3	Simply the Boolean function using **Boolean laws and theorems** and Implement the results using **logic gates.** i. x'y'z+x'yz+xy' ii. A'B'C'+ A'BC'+ AB'C'+ ABC'	05
2(a).	1	L3	2.4.1	Simplify the Boolean function in **SOP** and **POS** for the given function: **f(p,q,r,s)= Σm(0,1,2,5,8,9,10)** using **K-MAP.** Implement the results using **logic gates.**	10
2(b).	1	L3	2.2.3	A Boolean function of 3 input variables x,y,z produces an output of 1 whenever 2 or more input variables have the value 1 else the output is 0. Construct the Truth table and express the function in canonical SOP form and canonical POS form.	05
				OR	
3(a).	1	L3	2.4.1	Simplify the following using K-MAP in **POS** for the given Boolean function: f(A,B,C.D)= πM(0,1,2,3,4,5,7)+dc(8,9,11,12,13,15) Implement using Universal gates.	10
3(b).	1	L2	1.3.1	What is Universal gates? Realize Basic gates(AND, OR, NOT) using Universal gates (NAND and NOR).	05

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Course Outcome: At the end of the course the students will have the ability to	
CO - 1	Explain the fundamental concepts of Boolean algebra, Boolean theorems, K-MAP, Quine-McCluskey algorithm and apply the knowledge of theoretical foundations to simplify and solve Boolean functions.
CO - 2	Design and implement the combinational circuits.
CO - 3	Apply the knowledge of timing aspects and operation of Flip-Flops in conversion between different Flip Flops.
CO - 4	Illustrate the working of various registers and register circuits.
CO - 5	Design and implement synchronous/asynchronous counters and synchronous

Cognitive Domains	
L1	Knowledge
L2	Comprehension
L3	Application

PI's	
1.3.1	Apply engineering fundamentals
2.2.3	Identify existing solution/methods to solve the problem, including forming justified approximations and assumptions
2.4.1	Applies engineering mathematics to implement the solution.

— End —